

HDP – WORKING GROUP CARGO

OPERATIONAL CONCEPT DESCRIPTION

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2 INTRODUCTION

The purpose of this document is to consider the understanding of the user's needs and how the system shall operate to fulfill these needs and is to be used throughout the Cargoloop development. A user-oriented document that describes a system's operational characteristics from the end user's viewpoint.

3 EXISTING CARGO TRANSPORTATION SYSTEM AND OPERATIONS

3.1 SYSTEM

Simply defined, cargo transportation is the movement of goods from one area to another. Cargo transportation allows production and consumption to occur at different locations. Transportation is necessary for economic specialization. Freight transportation allows companies to (1) specialize in producing the products for which they are best suited and (2) trade with other companies to obtain products that can be made more efficiently by others.

Freight transportation can be considered from the perspectives of both supply and demand. Demand comes from businesses that need to move raw materials, supplies, and finished products. These businesses, called shippers, are the purchasers of freight transportation.

The supply of freight transportation is provided by the infrastructure and the companies that move the goods, called carriers. Freight infrastructure includes the roadway system, railroads, airports, marine ports, locks and dams on rivers, and pipelines. Freight carriers are the owners or operators of the trucks, trains, ships, and airplanes that provides transportation to shippers.

Other important private players in freight transportation include freight brokers, freight forwarders, and third-party logistics providers. Freight brokers assist shippers and carriers in assembling paperwork for international or complicated shipments. Freight forwarders consolidate multiple small shipments into larger shipments for transport. (This often involves preparing shipping and customs documents as well.) Third party logistics providers (3PLs) are companies employed to assume freight/logistics tasks previously performed in house by shippers.

The primary modes of freight transportation are truck, rail, marine, air, and pipeline. Each of these modes tends to provide different types of services and move different cargo types. One can think of freight transportation modes as providing a continuum of speed and service types (Figure 1). One end of the freight service continuum is characterized by fast and reliable delivery, but these high levels of service also cost the most. Air transportation is the most expensive and fastest.

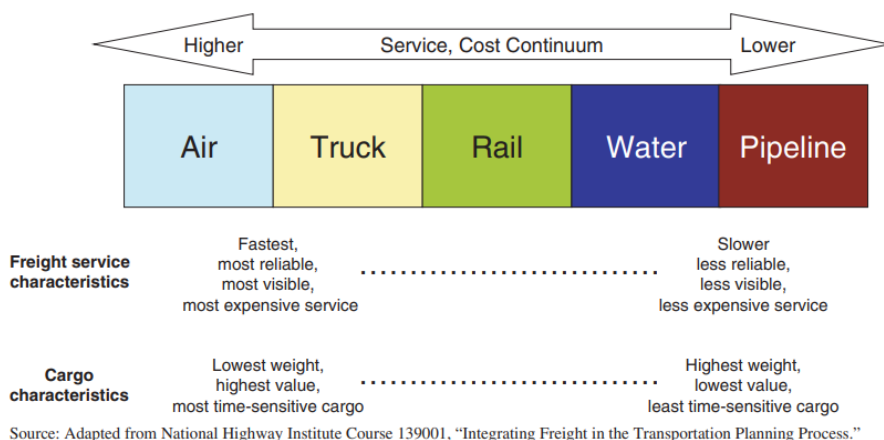


Figure 1: Service and cost continuum across cargo transport modes

Truck transportation provides rapid and flexible service for shippers, but at higher cost than rail transport. Marine and pipeline transportation are the least expensive in terms of cost per ton-mile, but they provide less rapid and flexible freight service.

Cargo characteristics determine the type of transportation service demanded by shippers. Companies shipping high value or perishable cargo tend to select truck or air transport to reduce transit time and gain higher levels of reliability. Air freight carries high-value goods for which delivery within a few hours is often critical, such as express parcels and fresh flowers.

Trucks move a range of products, but they move a greater percent of higher value commodities like finished consumer products, computers, and pharmaceuticals. Railroads tend to carry lower value, slow-moving bulk traffic (e.g., coal and grain), although they also move some higher value products (e.g., auto parts and finished vehicles). Domestic marine transport tends to carry low-value bulk cargo (e.g., coal and grain) for which speed does not matter. Pipelines are used primarily for petroleum products and natural gas. Overall, more expensive transportation services provide shippers greater visibility in terms of where their shipment is and when it will be delivered.

The length of haul is also an important shipment characteristic that determines mode choice. Trucks tend to capture a greater percentage of short-haul freight movements. Rail, marine, and air shipments tend to have a longer average shipment distance.

Freight shipments often use more than one mode of transportation. Trucks connect shippers to rail or marine transportation modes or provide the “last mile” of freight transportation to the customer. “Intermodal” freight typically refers to freight moving in containers or trailers that can easily be transferred between ships, railroads, and trucks. By reducing the cost of using multiple modes of transportation, intermodal freight movement allows shippers to use lower cost modes (such as rail or marine) for long-haul movements and then switch to truck carriers to reach a final destination.

3.2 PROCESSES

The transport processes considered relevant for Cargoloop operations are LTL (Less-than-full truck load) transport process and e-commerce transport process.

In LTL transport process (Figure 2), a truck picks up pallets at various customers in a region, transport that to a hub where the pallets are cross-docked and loaded into trucks to drive to the region of destination, where it is again cross-docked to allow efficient last mile (combing pallets from various suppliers and regions that go to the same final destination).

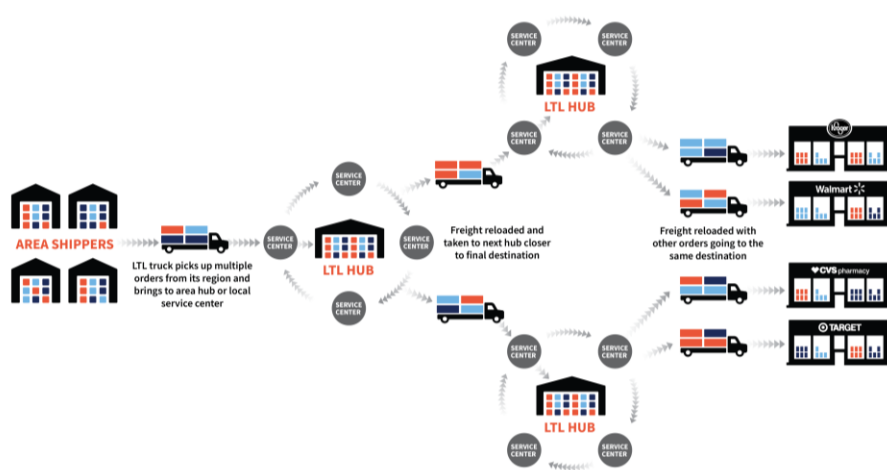


Figure 2: LTL transport process

For e-commerce shipments, the process on a high level is very similar to LTL with the main difference that a load unit is a parcel rather than a pallet and there are more steps in the process as there are more drop locations (Figure 3).

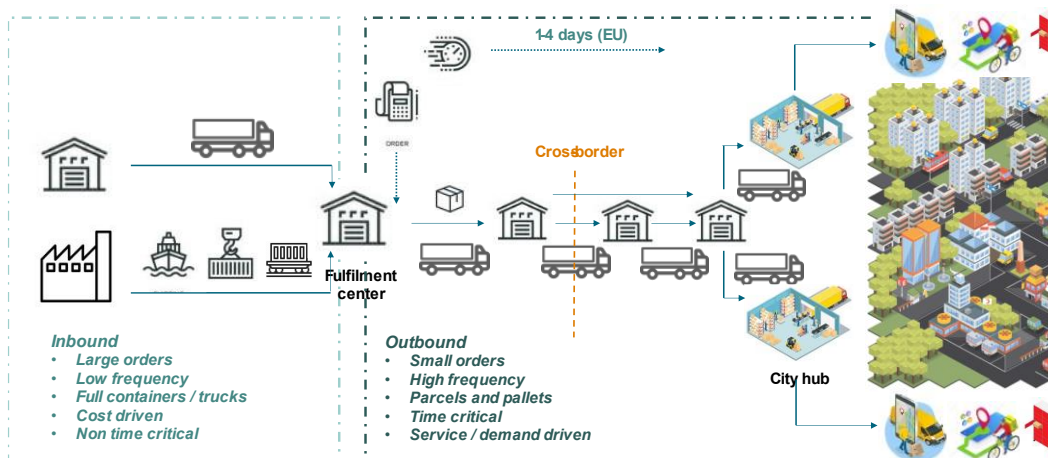


Figure 3: E-commerce transport process

3.3 OPERATIONS

Both LTL and e-commerce transport processes rely on a generic cross-docking transport process (Figure 4), including the first mile (delivery to cross-docking terminal), cross-docking process in the origin terminal, line-haul transit to the transfer terminal, cross-docking process at the transfer terminal, line-haul transit to the destination terminal and the last mile delivery to end customers.

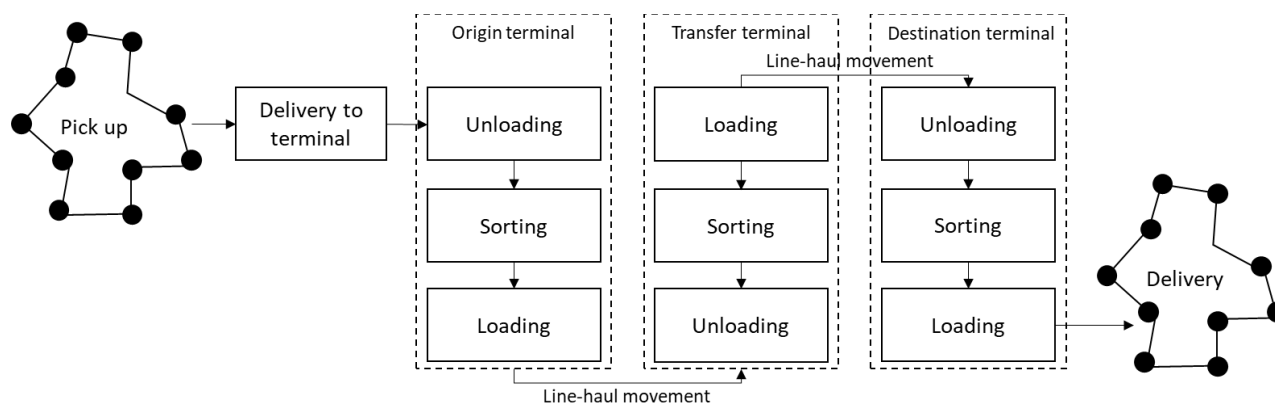


Figure 4: A generic cross-docking transport process

The cross-docking process includes unloading the trucks at unloading (inbound) docks and transporting it to staging lanes, sorting the cargo and loading it onto trucks in the loading (unbound) docks for the next stage of the cargo journey (Figure 4). The cross-dock facilities can have various configurations, an example of a single stage cross dock is presented on Figure 5.

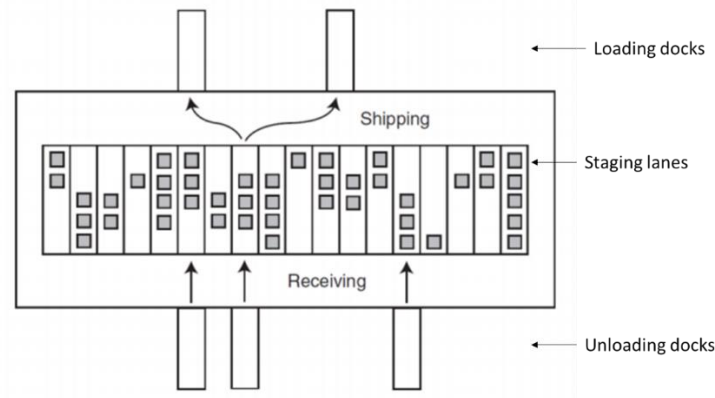


Figure 5: Single stage cross-dock, sorted on loading (outbound) docks

Under normal conditions, the road cargo transport operations are conducted during the day, with 10 peak hours between 6am and 4pm. The average share of daily traffic in a peak hour ranges from 6 to 8% (Figure 6).

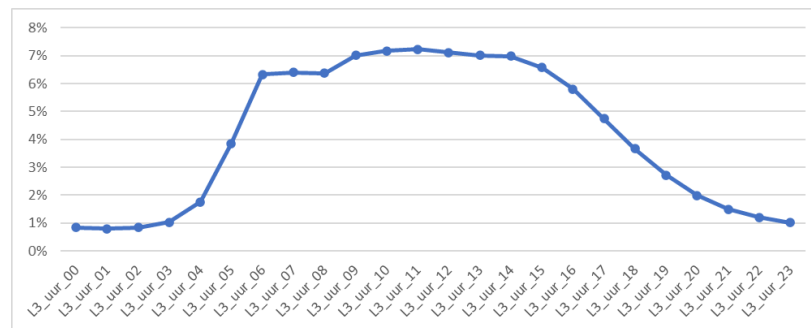


Figure 6: Cargo transport daily profile on main roads in Netherlands, 2020

The cross-docking operations are usually conducted over 24 hours, with three shifts of eight hours: a day shift (7:00-16:00), evening shift (16:00 – 23:00) and a night shift (23:00-7:00). The peak times for different types of cross docking operations (unloading, sorting, loading) differ. They can be concentrated on several moments a day or scattered throughout the day. If the arrivals are concentrated in the evening, then all arrivals are sorted in the night and then the night shift is operationally challenging in terms of inbound capacity and staffing. Because of age-related agreements there are fewer workers available in the night than a day and evening shifts. And also because there are no trucks leaving to the stores approximately between 21:00h and 3:00h there is a lot of outbound capacity needed to stage the pre produced routes. The day shift is busy (lots of trucks leaving and new routes produced) but also a lot of workers are available. The evening shift is special; the workers are available but usually there is limited production and a lot of staging lanes are empty, as this is the peak period for arrivals and then all arrivals sorted in the night shift.

In many cases the departure times are not restricted and the trucks depart when all the freight is loaded or unloaded. Inbound trucks that has restricted departure times must unloaded on time i.e. to prevent getting penalties by the inbound transport company. But in practice the trucks never leave before all freight is unloaded. However, outbound trucks in the e-commerce industry must leave exactly on time to meet the delivery appointments with customers. When some freight is delayed the truck will leave the cross dock without the complete freight in order to not delay all other parcel deliveries.

3.4 USERS

Cargo transport system has a number of internal and external users with whom it interacts during or as a result of its operations.

- Cargo owners and recipients are the end users of the system benefitting from its features and bearing the consequences of its failures.
- Logistics service provider. In many industries it is common that customer (cargo owner) works with one or more logistic service provider for the door-to-door service. This logistic service provider can outsource some of the activities to freight transport operators.
- Freight operators deliver the cargo transport service and are direct users of the transport infrastructure.
- Infrastructure providers are responsible for development and maintenance of the transport infrastructure.
- Governments and other authorities are responsible for planning and very often funding the transport infrastructure.
- Society is impacted by the operation of cargo transport system through its impact on economy, accessibility to products and environment.

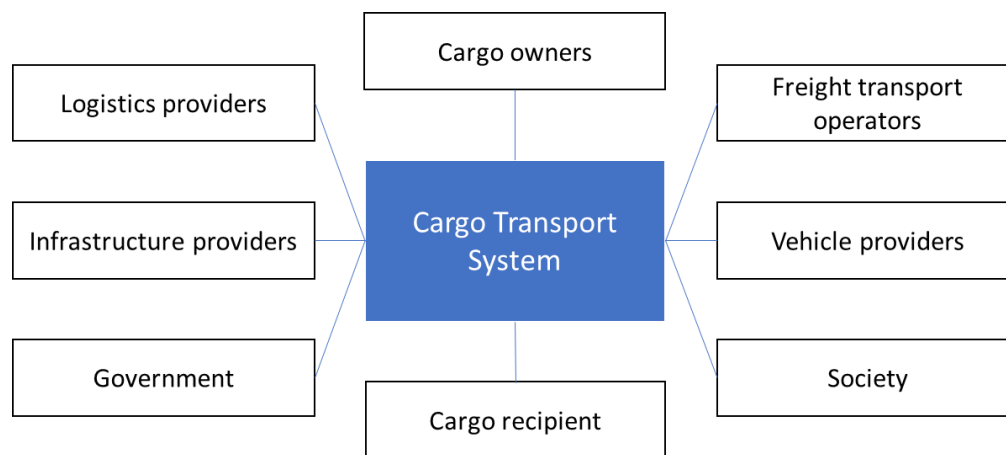


Figure 7: Cargo Transport System Users

3.5 LIMITATIONS AND CONSTRAINTS

Current continental supply and distribution networks are based on limitations of existing modalities, especially with regard to speed and environmental impacts (Figure 8).

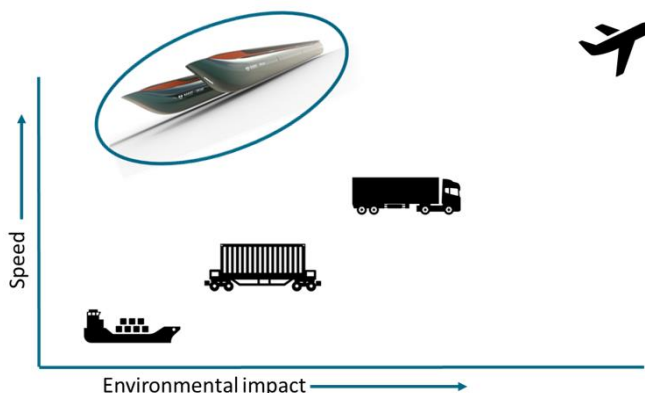


Figure 8: Speed and environmental impact across cargo transport modes

For example, road transport is highly flexible as road networks allow trucks to transport from anywhere to anywhere, but it also has major limitations.

- Road transport is the second largest source of carbon emissions in Europe. While freight transport accounts for about 27% of all transport related carbon emissions from fuel combustion¹, road freight transport is responsible for more than 70% of these emissions² which equals to about 6% of total EU carbon emissions. Furthermore, road freight volumes and related emissions have been steadily increasing over the last decade by about 1% annually³⁴. This is a major challenge to efforts to reduce emissions in line with the Paris Agreement and other global goals.
- Road freight transport volumes are expected to grow 55% by 2050 (with 2010 as a reference year). This growth is almost exclusively facilitated by heavy trucks. Total volumes of road freight transported annually in Europe range from 5 to 80 million tons for a motorway or express road, with an average of 15 million tons per road⁵.
- The EU aims to be carbon neutral by 2050. This cannot be achieved with current transport technologies even in the most optimistic scenario. Long distance transport is especially challenging to decarbonize. An electric light commercial vehicle becomes inefficient above 80 km/h, and only the maximum range of 150 km is possible to achieve with improved battery technologies. These vehicles are not suitable for long-haul transit. Electric heavy commercial vehicles are energy inefficient and the weight of required batteries is too high for long haul operations.

One of the key policies impacting logistics is a modal shift from road and aviation towards sustainable transport modes. In continental logistics, modal shift means shifting cargo from truck to rail or barge transport. This is difficult

¹ Greenhouse gas emissions from transport in Europe, European Commission, 2020

² https://ec.europa.eu/clima/policies/transport_en

³ <https://www.transportenvironment.org/newsroom/blog/eu-emissions-will-be-transport%E2%80%99s-last-lost-decade>

⁴ Road freight statistics, Eurostat

⁵ Calculations based on: Eurostat data 2019

to achieve, especially for outbound logistics⁶ as shipments are too small for a full truck load or a container and good would have to be first consolidated to full containers in order to utilize rail or barge transport. Rail and barge transport is not flexible enough to accommodate less than full truck/container loads.

The existing system also faces certain operational constraints - limitations placed on the operations of the current system such as:

- A constraint on the number of personnel available to operate the system (e.g. truck drivers). The EU sees a 36% driver shortage in 2020⁷. As more drivers are approaching retirement age, and the occupation of truck driver is not perceived an attractive one, this mismatch will become worse and seriously impact road freight capacity in the future.
- A constraint on the external influences (weather, traffic jams, ...)
- A constraint on the hours of operation, e.g. limited to access to secure terminals
- A constraint on environmental and socioeconomic efficiency. Also, to facilitate the expected growth in transport flows in the next decades, large investments in infrastructure are needed. As rail and barge are unfit for distribution networks, most of the growth will result in more truck and air movements. More freight on highways also means competition with personal transport and more congestion and accidents.

⁶ Shipping out of finished products to customers from a warehouse or distribution center

⁷ <https://www.iru.org/resources/newsroom/iru-report-forecasts-alarming-jump-driver-shortage-europe>

4 JUSTIFICATION FOR THE CARGOLOOP SYSTEM

The Cargoloop is expected to provide a viable alternative to air and truck transport, and to a limited extent to certain rail freight services. The most potential for Cargoloop is in distribution networks, the part of a supply chain in which ordered products are being delivered to customers. Typically, as soon as a customer orders a product, short lead times are a major factor of customer satisfaction and thus success of the manufacturer. Customers, both consumers as well as business are increasingly expecting lead times of 1 to a few days. An expectation that now can only be met if one uses an air-based express network or if the stock location is close enough to the customer to allow short lead times by truck.

As the shipments of small and medium cargo units are an emerging challenge for the logistics market which have not yet been fully addressed by the transport industry, the Cargoloop is envisaged as a complementary solution addressing this niche and releasing operational stress on the existing cargo transport modes. To do that Cargoloop responds to key challenges and needs of the logistics sector and their expectations towards the Cargoloop, which were identified through the market research activities.

- Cargoloop radically improves agility and efficiency of logistics operations for just-in-time and on-demand deliveries of small and medium sized shipments and packages, while minimizing their costs and carbon footprint
- Cargoloop unlocks new market opportunities and economies of scale for time-sensitive and demand-sensitive products (reducing market access times, increasing geographical coverage, enabling new types of services and products)
- Cargoloop enables cost-effective transition to digital and automated logistics operations to increase responsiveness to demand volatility and changing customer requirements (logistics as a service, automated fulfilment, packaging modularization, physical internet)
- Cargoloop reduces carbon emissions from the transport and logistics activities and allows the industry to comply with stringent environmental standards.
- Cargoloop also creates opportunities for supply chain optimization and automation, increasing the coverage of existing logistics facilities, and reducing the need for capital investments in new or expanded facilities.

Also, the Cargoloop responds to the needs of the logistics industry related to the quality of service, identified through the market research activities (Figure 9).

- Reliability and punctuality of service. The Cargoloop is well placed to respond to this need, as its enclosed environment and separation from other transport infrastructure eliminates the risks of congestion and reduces the risks of external disruptions, especially adverse weather conditions which can cause disruptions to all other transport modes.
- Short transit times. Cargoloop enables transit times of a fraction of the time offered by any other land transport mode.
- High quality of cargo handling. High level of automation in the Cargoloop system allows for better control and management of cargo handling and reduces the risks of accidental cargo damage.
- Low costs. Lean infrastructure of the Cargoloop and low maintenance costs allow for the transport costs being lower than any comparable transport mode.
- High frequency and availability of services. Cargoloop has a potential for multiple shipments per minute and continuous operations up to 24/7
- Low carbon emissions. Cargoloop is a fully electric transport mode which has a potential for being powered from renewable sources.

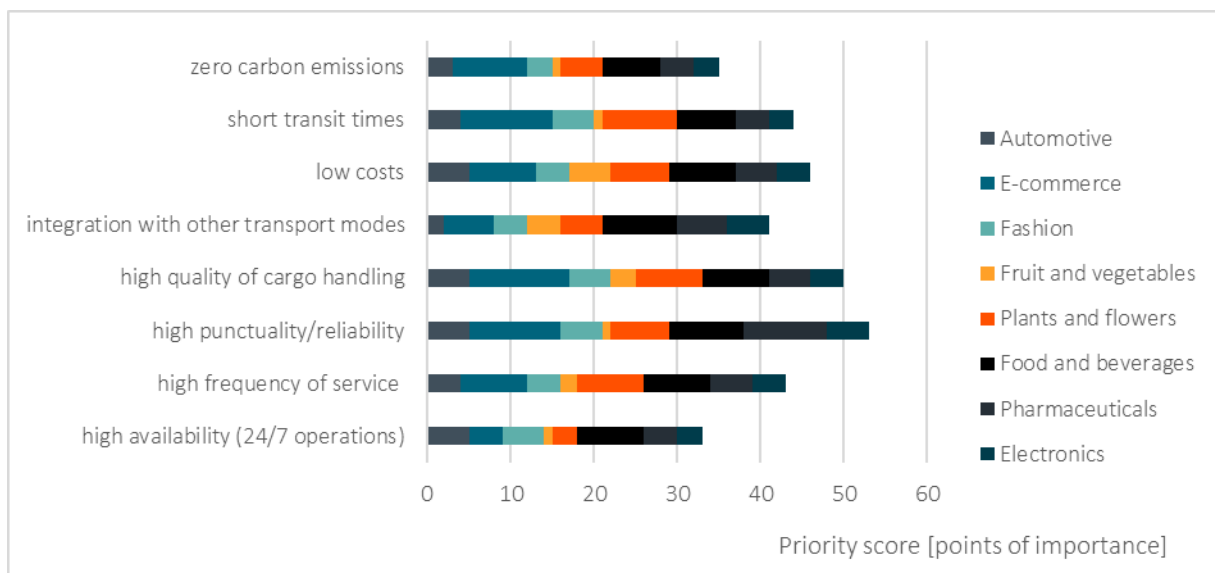


Figure 9: Industry needs for the quality of service

Benefits of the Cargoloop do not end on advantages experienced by its users and realizing the Cargoloop is also expected to bring a range of positive socioeconomic and environmental impacts.

- Relieving congestion, improving safety and reducing the maintenance needs of existing transport networks by providing a new transport solution for moving existing and future freight volumes,
- Improving efficiency of public infrastructure investments by providing viable cost-effective alternative to expansion of the existing transport networks,
- Improving livability of densely populated areas by introducing a new transport infrastructure which due to its size minimizes the spatial footprint necessary to increase the capacity of the whole transport system,
- Improving environmental performance of transportation by providing an emission free and energy efficient cargo transport solution,
- Accelerating innovation by enabling faster uptake of automation and digitalization,
- Increasing productivity and resiliency of the economy by enabling faster transit of products between manufacturing facilities and the market.

5 PROPOSED CARGOLOOP SYSTEM

5.1 SYSTEM USERS

Cargoloop system has a number of internal and external users with whom it interacts during or as a result of its operations. As the Cargoloop system is a component of the overall cargo transport system, its users fall into the same general cargo transport system user categories⁸. Each user category has different needs, expectations and objectives towards the Cargoloop system.

- Cargo owners primary expectation is that their cargo is delivered unharmed, quickly and on time, and that they are provided with information about the cargo status and location (preferably real time data).
- Logistics service providers focus on reliability of the system (timely and error free services), low costs of transport operations and accessibility of the service (such as connectivity with other transport modes, availability of capacity slots for the cargo they would like to move).
- Freight operators interact with the Cargoloop system through its hubs, providing the first and last mile transport services. Their primary concern is accessibility of the hubs and reliability of the time slots allocated to them to (un)load their vehicles.
- Governments and public agencies are focused on the impacts of the Cargoloop system on economy, safety, security and environment.
- Society is impacted by the operation of the Cargoloop transport system through its impact on economy, accessibility to products and environment.
- Internal users of the Cargoloop system include cargo handling, communication and control, and maintenance personnel. The common denominator of their needs is safety of the system and information and data which they require to efficiently perform their duties with low error rates.

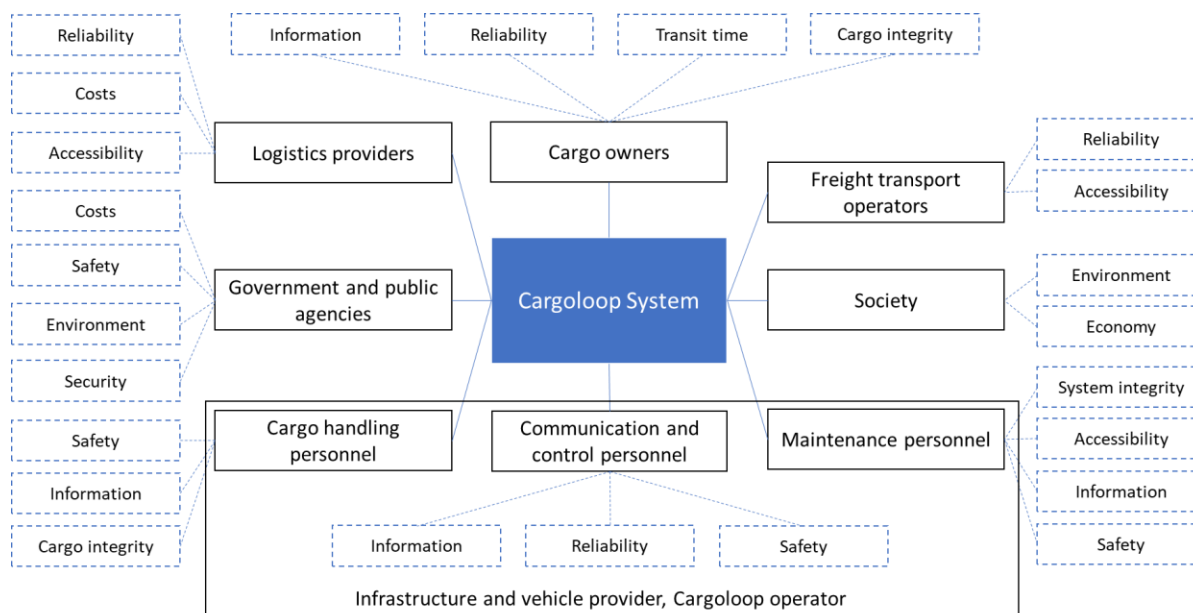


Figure 10: Cargoloop users and their priorities

⁸ In case of the Cargoloop, the infrastructure provider, vehicle provider and Cargoloop system operator are consolidated and regrouped into internal Cargoloop system personnel.

Special types of personnel are needed to operate a Cargoloop hub and network. They must be able to continuously operate at very low error rates. Goods move so rapidly through the facility that an error will often be detected only when it has already reached the customer. When that occurs, all that can be done is to correct processes to reduce the potential for future errors – damage caused by an error missed often cannot be rectified.

5.2 SYSTEM GOALS AND OBJECTIVES

The Cargoloop service vision defines the goals and objectives for the Cargoloop system through the range of services offered to its users:

- Moving small and medium, standardized, and configurable shipment units
- Short transit times enabling same-day delivery within any 2,000 km distance
- On-demand capacity
- Seamless interfaces with other transport networks and last/first mile services
- Fully traceable and continuous operations in controlled environment
- Lean infrastructure enabling easy integration into production, distribution and demand centres
- Lower costs and emission levels than any other comparable transport option

5.3 SYSTEM OPERATIONS

The Cargoloop system operates as a part of the logistics environment and is integrated with other transport modes (predominantly road transport) to form a delivery process (Figure 11). The delivery process starts with picking up cargo from different locations and delivering it by truck to the Cargoloop departure hub where it is unloaded, sorted and loaded onto Cargoloop vehicles. The Cargoloop vehicles transport cargo to the arrival hub where the reverse process takes place (unloading Cargoloop vehicles, sorting and loading trucks) and the cargo is delivered by trucks to its final destinations.

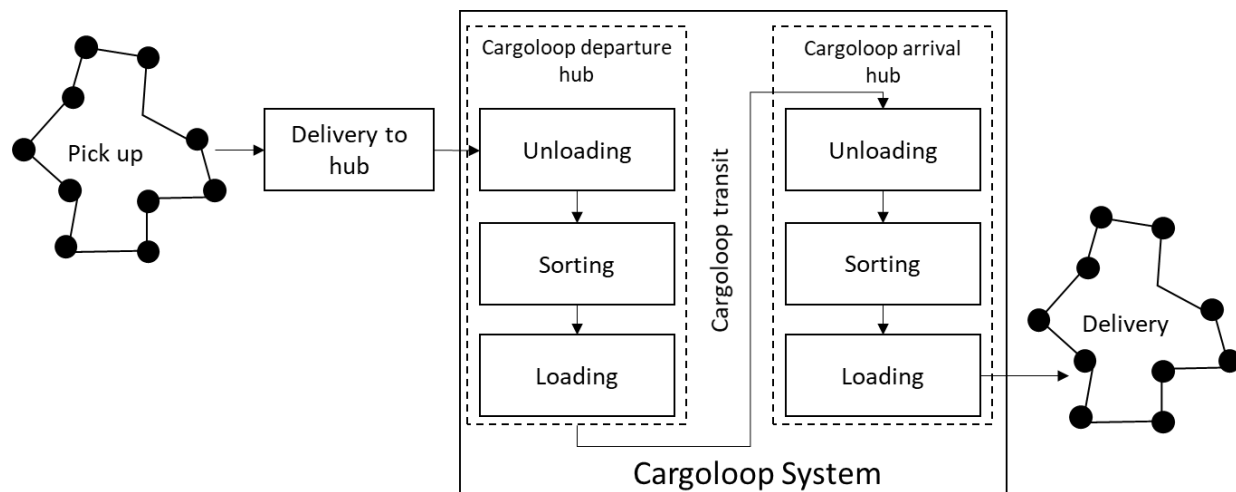


Figure 11: Cargoloop system operations in the delivery process

The Cargoloop system allows for different service types: express, standard (fast) and overnight service.

- The express service has the highest priority in processing through the Cargoloop hubs and the Cargoloop network. The cargo is not buffered but goes directly from the inbound truck to the Cargoloop vehicle at the departure hub (Figure 12) and directly from the Cargoloop vehicle to the truck at the arrival hub (Figure 13).

- The standard (fast) service has the second priority. The cargo is buffered in the Cargoloop hubs awaiting to be combined with other pallets transported to the same destination, and for a route to the arrival hub to become available and ready (Figure 12). At the arrival hub, the reversed process is applied: the cargo is buffered and sorted awaiting for the truck to pick it up and deliver to its final destination (Figure 13).
- The overnight service is arranged to provide off-peak or overnight delivery. The cargo is buffered in the Cargoloop hubs awaiting for the off-peak or overnight delivery window, when it is sorted, combined with other pallets and transported to the arrival hub (Figure 12). At the arrival hub the cargo is buffered, sorted and awaits the morning truck to pick it up and deliver to its final destination (Figure 13).

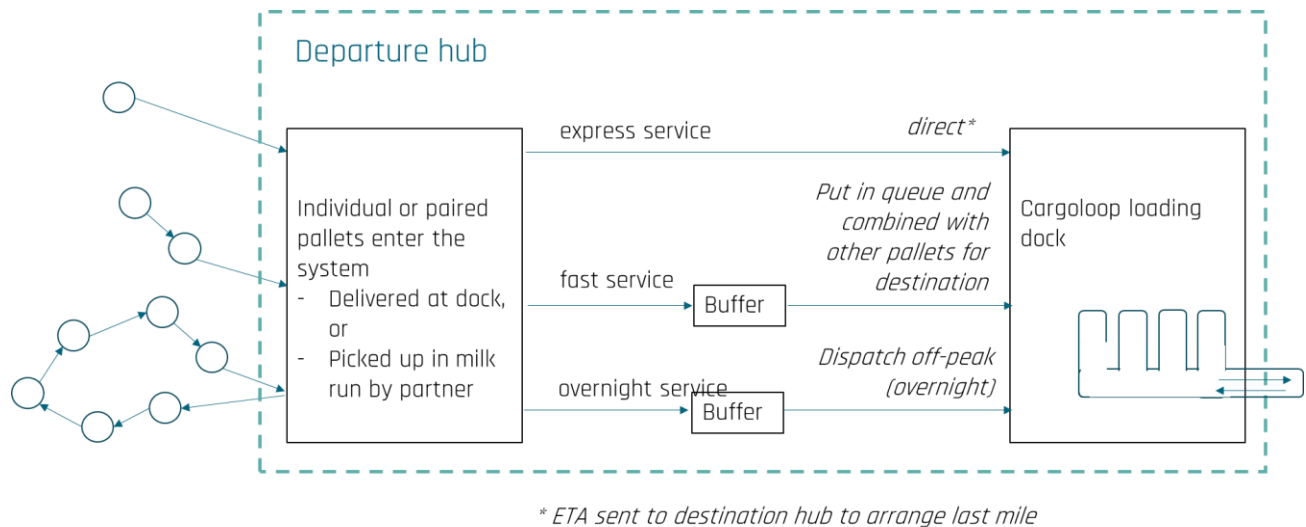


Figure 12: Cargoloop hub departure operations

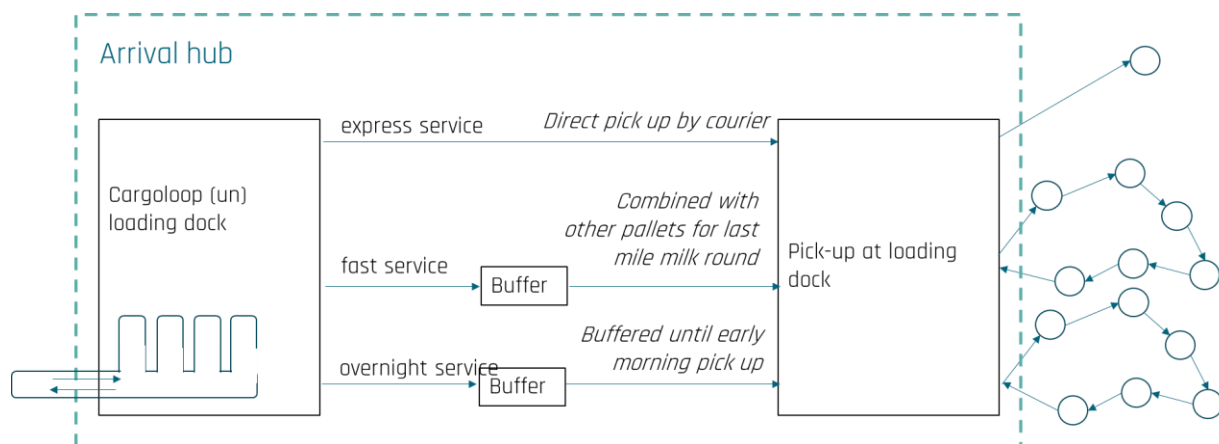


Figure 13: Cargoloop hub arrival operations

Processing the cargo through the hub can be deconstructed further into specific operations. All Cargoloop operations take place within 24 hours.

- **Cargo unloading from a truck.** Products arriving at the Cargoloop hub by truck from different destinations are unloaded from the inbound trailers in the receiving docks (Figure 14). Depending on the size of the items to be unloaded and the way they are packaged, the operation can be done manually, with

a hand pallet truck, a powered pallet truck or a forklift (Figure 15), or automatically (e.g. robotic arm). Bulk goods might need to be palletized while being unloaded, and multireference loads must be sorted.

- **Cargo check-in.** After unloading from a truck, the cargo is subsequently transported by a conveyor to a check-in gate. The check-in process aims to verify the integrity of the data regarding the incoming cargo, by checking cargo identity and destination, and whether the cargo complies with the dimension and weight requirements of Cargoloop system. The control consists of weighting the products, visually checking or scanning the product dimensions, and verifying the data from cargo labels and tags with the data from the Cargoloop system database. All the non-compliant cargo items are considered as exceptions and processed separately.
- **Buffering, sorting and staging for departure.** The cargo which is successfully checked-in (excluding exceptions) is directed, depending on the service type, towards the express bypass (express service), the Cargoloop staging area (standard service) or the overnight buffer area (overnight service). The cargo directed to the Cargoloop staging area is sorted by destination, placed in dedicated staging lanes and awaits for the Cargoloop route to become available.
- **Loading onto the Cargoloop vehicles.** When the Cargoloop route becomes available, the cargo is transported to the Cargoloop platform and loaded into the Cargoloop loading bay. The loading bay awaits on a platform for the Cargoloop vehicle to arrive and is loaded into the vehicle.
- **Unloading from the Cargoloop vehicles.** When a Cargoloop vehicle arrives at the hub, the loading bay is removed from the vehicle into the Cargoloop platform. Cargo is unloaded from the loading bays on the Cargoloop platforms and transported by a conveyor to a check-out gate.
- **Cargo check-out.** At the check-out gate, the cargo ID and final destination is verified and the cargo is directed to a dedicated staging lane for outbound shipping.
- **Cargo loading onto a truck.** The cargo is loaded onto trucks from a dedicated staging line in the shipping docks. The loading process is either manual or automatic.

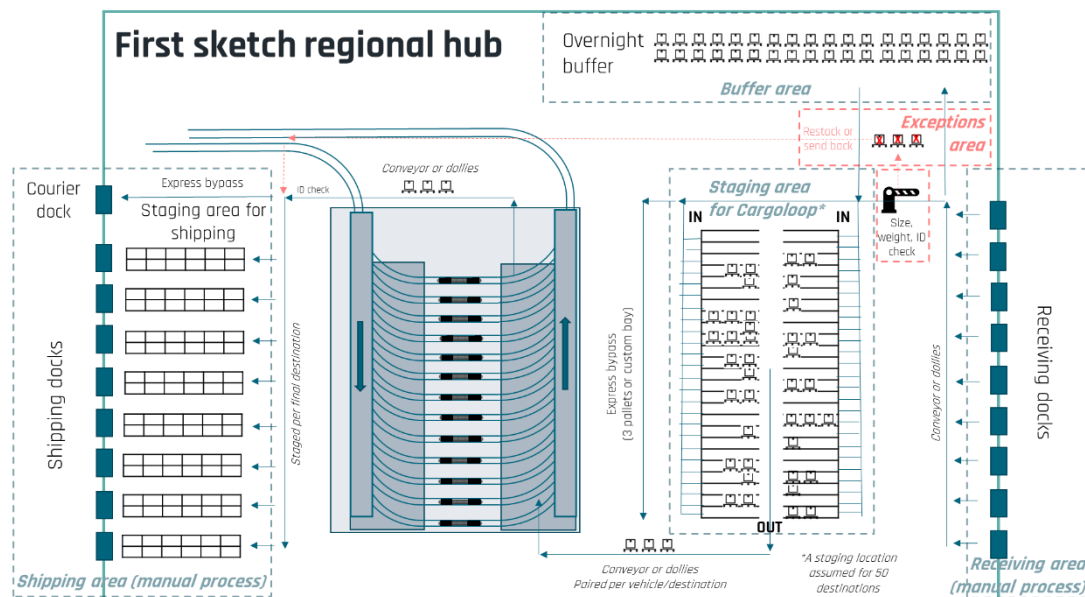


Figure 14: Cargoloop hub operational components and functionalities

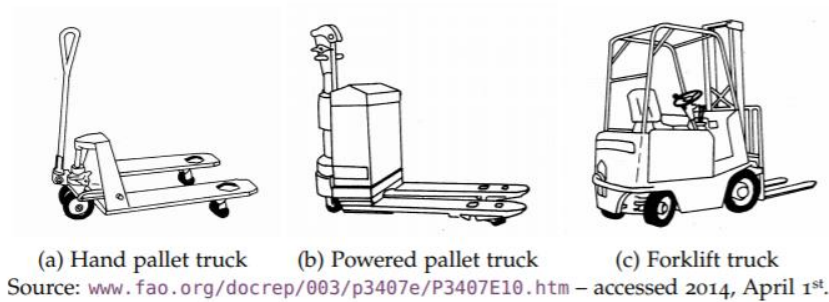


Figure 15: Examples of manual (un)loading equipment

The Cargoloop network comprises the core network, regional networks, and local connections.

- Core network includes the highest priority strategic corridors linking the most important long-distance metropolitan areas at high-speed. Generally the core network itself doesn't connect to any hubs directly, but branches off into regional networks.
- Regional networks directly link local hubs and feed into the core network.
- Feeder (or 'Local') links generally have the same specifications as the regional network, except that they consist of a single tube with means to buffer vehicles. These links are substantially lower cost than regular double-tube links.

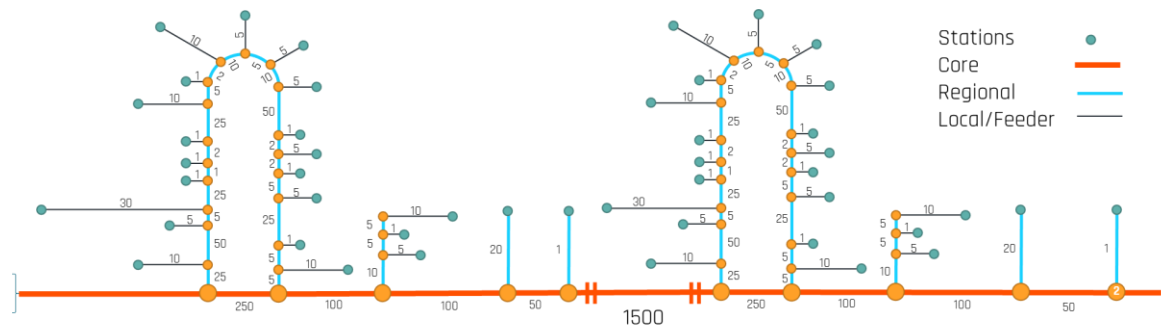


Figure 16: Example network showing the different types of elements within the comprehensive network. The numbers are the distances in km. Note that the network has been compressed at the 1500 km link, and in reality there would be many more regional network branches within the compressed section 1500 km, typically in a ratio of (regional+local/feeder):(core) of 5:1.

The operation of the network is also conducted on multiple levels which represent the different aspects of the network and vehicle control (Figure 17).

- **Network level:** Decides the destination of each Cargoloop vehicle within the network. The operating system has the capabilities of predictive, adaptive, and smart scheduling strategies in the form of optimization algorithms to minimize delays and power consumption as well as improve throughput. It also mitigates network level demand variations (daily, monthly, seasonal, etc), capacity limitation (saturation of specific links), disturbances (pod dispatching problems, temporary system malfunction, etc.), disruptions (planned or unplanned maintenance, emergency shutdowns of sections, etc), and delays (pod delays, structural delays, etc). Communicates to (regional), local and pod level.
- **Regional level:** This level is optional depending on the size of the network and the achievable network level operating frequency. The regional level has similar responsibilities as the network level OS but details

the scheduling real-time for a smaller portion of the network if the network level can't manage the size and required frequency. It has to communicate changes both downstream (local and vehicles) and upstream (network). It knows the location of all vehicles in the system.

- **Local level:** Translates the network or regional level orders to low-level stations systems and propulsion systems along the main infrastructure and on/off ramps. It also receives updates from these systems on vehicle position and station status. By combining this information, it ensures ensuring safe separation between pods. The low-level systems such as the propulsion drives and station airdock actuators execute the work. The local level also monitors and reports the system performance and safety status back to network level or vehicles.
- **Vehicle level:** The vehicle operating system has the functionality of optimizing its own headway (through on-board propulsion systems) and executing safety protocols (through the emergency braking system). Moreover, it has the role of taking switches and controlling its direction of travel based on network or regional level orders.

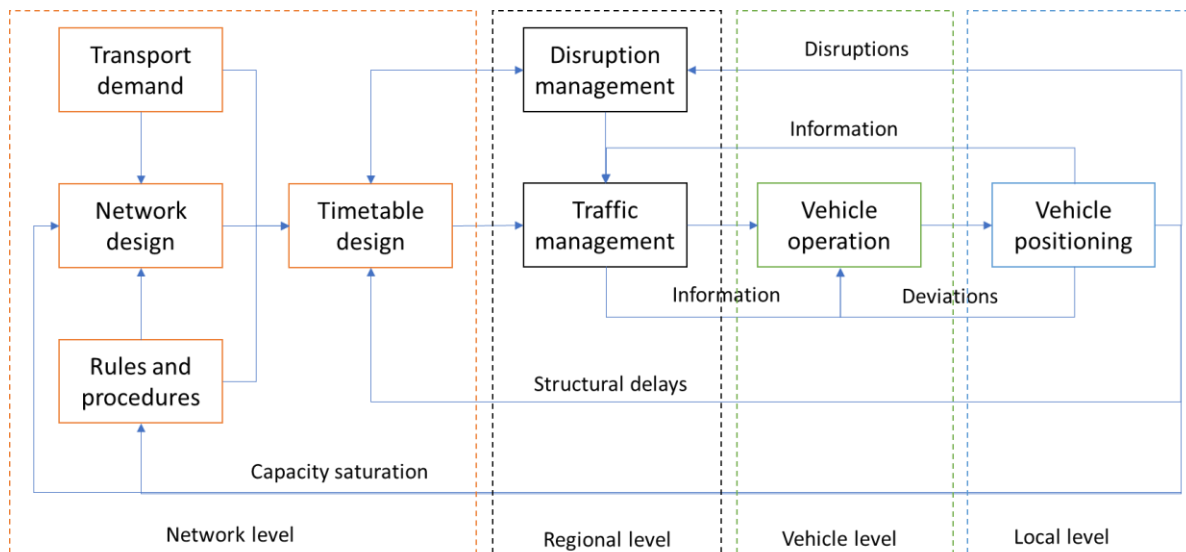


Figure 17: Cargolop network operations

5.4 SYSTEM CAPABILITIES

Cargoloop offers the following capabilities:

- The Cargoloop service is realized with hyperloop technology - an electrically powered mode of land transportation with vehicles travelling through a network of low-pressure tubes.
- Short transit times enabling same-day delivery within any 2,000 km distance with the cruising speed of up to 500-700 km/h
- Autonomous operations with the ability for direct point-to-point transit of each shipment without intermediate handling
- High accessibility of the system to customers, with the Cargoloop network designed to be accessible within 30 minutes driving distance from any major logistics location.
- Cargo envelope dimensions of 1.35 x 1.35 x 1.65+ (width x height x length). Vehicles with capability of transporting shipment sizes of ~3 half-pallets in length
- High capacity of 20,000 half-pallets per hour per direction
- Seamless interfaces with other transport networks and last/first mile services
- Lean infrastructure enabling easy integration into production, distribution and demand centres. Small tube size enabling cargo envelope, (un)loading bay not larger than a truck (un)loading platform (Figure 18).
- Lower costs and emission levels than any other comparable transport option
- Fully traceable and continuous operations in enclosed environment providing stable and controlled transit conditions and protection from external factors (weather, temperature, etc)

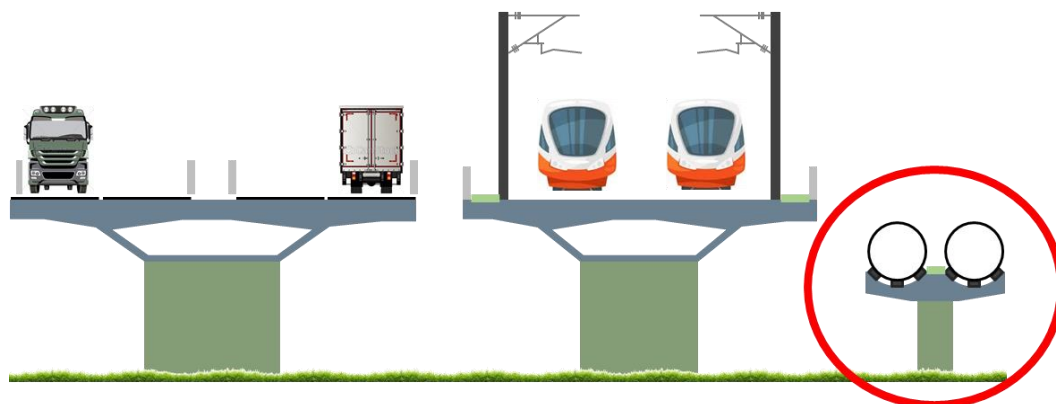


Figure 18: Cargoloop infrastructure – spatial comparison with other transport modes

Cargoloop system capabilities reflect goals and objectives of the system and user needs and expectations.

- High reliability of service and operations is realized by:
 - Enclosed environment of the tube, protecting the system from external disruptions such as adverse weather
 - Continuous 24/7 system operations
 - Automated operations reducing the error rate of the system
 - Direct point-to-point transit of each shipment without intermediate handling
- Short transit time is realized by:
 - High cruising speed
 - Direct point-to-point transit of each shipment which shortens the transit time by removing intermediate handling points

- Low service and infrastructure costs
 - Lean infrastructure scalable to adjust to demand levels
 - Small tube size enabling cargo envelope, (un)loading bay not larger than a truck (un)loading platform (Figure 18).
 - Energy efficient low maintenance operations
- Accessibility of service and infrastructure
 - High capacity of the system
 - Cargoloop network accessible within 30 minutes driving distance from any major logistics location.
 - Lean infrastructure enabling easy integration into production, distribution and demand centres.
 - Small cargo envelope dimensions of 1.35 x 1.35 x 1.65+ (width x height x length) and vehicles with capability of transporting shipment sizes of ~3 half-pallets in length, allowing for accessibility of the system service by customers with smaller shipments (LTL, parcels)
- Information
 - Fully traceable operations
 - Automated operations allow for high level of data available in the system
- Cargo integrity
 - Enclosed environment of the tube, providing stable and controlled transit conditions and protection of cargo from external factors (weather, temperature, etc)
 - Automated operations reducing the error rate of the system
 - Ability for direct point-to-point transit of each shipment without intermediate handling
- Safety and security
 - Autonomous operations
 - Monitoring systems
- Environment
 - Small infrastructure footprint (Figure 18).
 - An electrically powered and energy efficient operations

5.5 OPERATING ENVIRONMENT MODIFICATIONS

Although the intent is for the Cargoloop system to seamlessly integrate into the existing operating environment as much as possible, in order to take full advantage of certain Cargoloop capabilities, the users will need to make some changes to make the best use of the Cargoloop system.

- The shipments intended to be transported with the Cargoloop system need to be arranged into half-pallets or other shipment units with dimensions fitting the cargo envelope of the Cargoloop system
- For the shippers who currently use the door-to-door road transport system, switching to Cargoloop would require additional first and last mile services.
- The companies wishing to integrate Cargoloop hubs with their existing facilities would need to define the level of automation they are willing to adopt for the Cargoloop hub
- Lean Cargoloop hub infrastructure and small footprint requires that the first and last mile services are conducted in a timely fashion to reduce the time their vehicles spend at the Cargoloop hub
- Cargoloop corridors with high density traffic (core network) will require reinforcement of the electric grid by utility companies to accommodate the electricity needs of the system
- Emergency response procedures need to be developed to reflect specifics of the Cargoloop system
- Land use policies may need to be reviewed to allow Cargoloop infrastructure to be developed in urban areas

In a practical example of the operational environment (Figure 19), the Royal Flora Holland would need to review its packaging policies to make sure that the shipments fit the cargo envelope of the Cargoloop system. They would also need to define the level of automation of the Cargoloop hub to ensure best integration with their operations. Unless the departure and arrival hubs are fully integrated with their warehousing facilities, they would also need to consider the first and/or last mile services to complement the Cargoloop operations.

Also, the utility companies would need to be engaged at the infrastructure development stage (feasibility study and design phase) to assess whether there is a need for electric grid reinforcement.

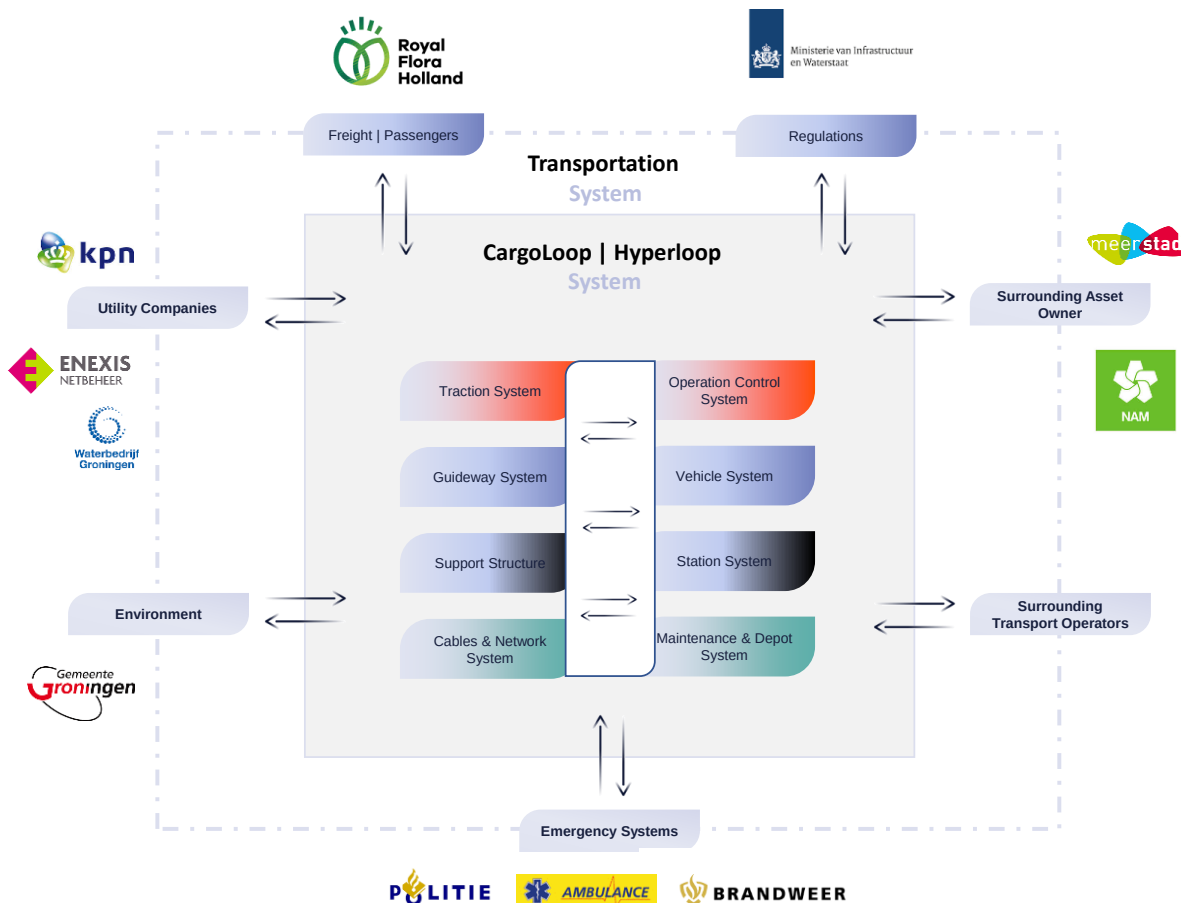


Figure 19: Practical example of the operational environment of the Cargoloop system

6 ANALYSIS OF THE PROPOSED SYSTEM

Implementation of the Cargoloop system brings numerous benefits to its customers:

- Speeding up the flow of products which will result in more frequent deliveries and reduced inventory obsolescence
- Cost reductions (warehousing costs, inventory-holding costs, handling costs, transportation costs, labour costs)
- Improved resource utilization, a better match between shipment quantities and actual demand and better control of the distribution operations
- Reduction of carbon emissions from the transport and logistics activities and allows the industry to comply with stringent environmental standards.
- Reducing market access times, increasing geographical coverage, enabling new types of services and products
- Increasing the coverage of existing logistics facilities, and reducing the need for capital investments in new or expanded facilities.

Disadvantages and limitations of the proposed system from the customer point of view are primarily related to the changes in operational patterns that they need to introduce in order to fully utilise the Cargoloop system, as described in the previous section, especially to shift towards smaller, half-pallet sized shipments.

